

Comparison of TetrAMM, FX4, and SRS570 + Quantum V2F100 + scaler

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Motivation

Many APS beamlines measure ion chamber and photodiode currents using the following counting chain.

- Stanford Research Systems SRS570 current amplifier. This outputs a -5V to +5 V signal proportional to the input current. The range (A/V) and analog filtering can be adjusted from the front panel or from EPICS.
- Voltage to frequency converter. Commonly used models are:
 - o Nova N101VTF 4-channel system. 0-10V input, switch-selectable 1,2,5, and 10 MHz/V output. These were purchased in the early days of the APS and have not been available for many years.
 - o Quantum VTF100 2-channel system. Input: 1, 2.5, 5, or 10V max. Output: 10, 25, 50, or 100 MHz. This has recently been discontinued and is no longer available. This is the model that was tested in this report.
- Scaler or multi-channel scaler. Models commonly used include the Joerger VSC8, SIS3820, and Measurement Computing USB-CTR08. The USB-CTR08 was used for this report.

While these counting systems have been very useful, they are becoming obsolete, particularly because of the lack of V/F converter module suppliers. They are also much more expensive per channel than the quad electrometer solutions now available. These include the following units that were tested in this report.

- TetrAMM from CAENels. This digitizes 4 currents at 100 kHz. 2 selectable current ranges are specified at the time of order. The unit used for these tests has ranges of ± 120 nA and ± 120 μ A. The current readings can be averaged on-board the device, and additionally in the EPICS quadEM driver.
- FX4. This digitizes 4 currents at 100 kHz. 8 current ranges from ± 120 nA to ± 10 mA are available on all units. The current readings can be averaged on-board the device, and additionally in the EPICS quadEM driver.

The TetrAMM has been widely adopted for measuring beam position monitors in the APS storage ring and on many beamlines. However, there appears to be only anecdotal evidence on whether the TetrAMM or FX4 have similar performance compared to the traditional SR570 counting chain for experimental data acquisition with ion chambers and

photodiodes. This report was done to provide a quantitative comparison under controlled and reproducible conditions.

Summary of the Results

The measurements were done with a computer-controlled current source and reading the measured currents from each system. For each device 3 tests were performed for each of 2 current ranges (6 tests total). Each test measured 1000 points with 0.1 seconds per point. The current ranges were 0 to 100 μA and 0 to 100 nA.

1. Constant current of 1% of range maximum to measure noise over 100 seconds at 10 Hz.
The results below are the standard deviations in nA or pA.

Current range	SR570	TetrAMM	FX4
100 μA	0.86 nA	0.20 nA	0.03 nA
100 nA	0.97 pA	1.86 pA	1.73 nA

2. Linear ramp of the current from 0 to range maximum. The results below are the standard deviations from the best fit straight line in nA or pA.

Current range	SR570	TetrAMM	FX4
100 μA	6.1 nA	2.5 nA	4.4 nA
100 nA	13.1 pA	11.1 pA	11.9 pA

3. Single-cycle sine wave with total range of 1% to 99% of range maximum. The results below are the standard deviations from the best site sine wave in nA or pA.

Current range	SR570	TetrAMM	FX4
100 μA	5.2 nA	2.5 nA	4.4 nA
100 nA	12.4 pA	12.2 pA	10.7 pA

The details of the test configuration and the individual results form the rest of this report.

Current source

The current was generated using a 16-bit digital to analog converter (DAC) in a Measurement Computing USB-1808X. The voltage output passed through either a 100 k Ω or 10 M Ω resistor and then to the current input of the device under test. The 100 k Ω resistor produces 100 μ A with a 10 volt output from the DAC, while the 10 M Ω resistor produces 100 nA with a 1 volt output. The DAC was driven in the “waveform generator” mode in the USB-1808X. The 1000-point output voltage waveform is streamed from the USB-1808X at a precise 10 Hz rate, so there is no jitter in the time between output values. The DAC output was also connected to an 18-bit analog input (A/D) on the USB-1808X. This provides a readback of the actual output voltage with better resolution than the DAC. The A/D was run in the the “waveform digitizer” mode in the USB-1808X, so it is reading the voltage input at precisely 10 Hz, sampling each output from the DAC.

The following screenshot shows the configuration of the USB-1808X when generating the sine wave with 100 μ A.

The screenshot displays the USB-1808 module configuration software interface. The window title is "USB1808_module.adl@corvette". The main title bar reads "USB-1808 13MCBOX:USB1808:". The interface is divided into several sections:

- Model Information:** Model USB-1808X, Unique ID 021515D8, UL version 1.2.0, Model # 318, Firmware 1.03, Driver version 4.4, Poll sleep time (ms) 50.0, Poll cycle time (ms) 50.6.
- Waveform generator:** Trigger Internal, Retrigger Disable, Trigger count 1, Clock Internal, Repeat One-shot, Current point 1000, # points 1000, Frequency 0.0100, Time/point 0.1000, Total time 100.0000.
- Trigger:** Mode Positive edge.
- Digital I/O:** Inputs 1, 2, 3, 4 (all Low), Outputs 1, 2, 3, 4 (all Low), Direction In, In, In, In.
- Output 1:** Enable Enable, Waveform Sin wave, Amplitude 9.8000, Offset 5.0000, Pulse width 1.0000e-04, Pulse delay 0.0000.
- Output 2:** Enable Disable, Waveform Pulse, Amplitude 0.2000, Offset 0.1000, Pulse width 0.0500, Pulse delay 0.0000.
- User-defined waveforms:** Frequency 0.0000, Time/point inf, # points 1.
- Pre-defined waveforms:** Frequency 0.0100, Time/point 0.1000, # points 1000.
- Analog input:** 8 channels, each with a value (e.g., 4.9951, 0.0350, -3.8605, etc.) and a 1 second Read button.
- Waveform digitizer:** Current point 61, # points 1000, Time/point 0.1000, Total time 100.0000, # points 1000, First chan 1, # chans 1, Burst mode Disable, Trigger Internal, Retrigger Disable, Trigger count 1, Clock Internal, Continuous One-shot, Auto restart Disable, Read rate 1 second, Read button.
- Analog output:** 2 channels, each with a value (e.g., 5.0000, 0.0000) and a 1.0000 Read button.
- Counters:** 2 counters, each with a value (e.g., 4, 5) and a Reset button.
- Encoders:** 2 encoders, each with a value (e.g., -327, -2) and a Reset button.
- Pulse generators:** 2 generators, each with Frequency, Period, Duty Cycle, Width, Delay, Pulses, Idle State, and Start/Stop buttons.

FX4 configuration

The following screenshot shows the configuration of the FX for the tests on the 0-100 nA current range.

quadEM.adl@corvette

Quad Electrometer (QE1:FX4:)

Model FX4

Firmware 26.06.181604

Sample time 1.0000e-03

Acquire Done Acquire

Acquire mode Continuous Continuous

Range 100 nA slow 100 nA slo

Weights Geometry Diamond Diamond

Values per reading 100 100

Averaging time 0.10 0.10

Fast averaging scan I/O Intr

Fast averaging time 0.100 0.100

Acquisitions 1 1

Acquisitions done 3030429

Read data Read

To average 100

Averaged 100

To average fast 100

Current units nA nA

Trigger mode Free run Free run

Trigger polarity Positive Positive

Bias voltage 0.00 0.00

Voltage readback -2.48

Bias state Off Off

Bias state readback Off

Time series

Acquire Erase/Start Stop Acquiring Plugin control

Time domain plots Frequency domain plots

X combined X combined

Y combined Y combined

Individual Individual

Status

Ring overflows 0

Status rate Passive Asyn record

Reset Reset

Plugins

Attributes

File

Macros

Status File not found

Buffers

Pre-allocate 100 Allocate

Used/Alloc/Free 69 191 122

Memory max/used (MB) 0.0 4.4

Buffer & memory polling 1 second

Empty free list Empty

Signal	1	2	3	4
Current	82.187	0	-0	0.001
(Fast)	82.058	0	-0	0.001
(Sigma)	0.171	0	0	0.001
Offset	0.000	0	0	0.000
Scale	1.000e+00	1.000e+00	1.000e+00	1.000e+00
Precision	3	0	0	3
	X	Y	All	
Sum	82.412	-0	82	
(Fast)	82.295	-0	82	
(Sigma)	0.162	0	0	
Difference	-82.410	0		
(Fast)	-82.293	0		
(Sigma)	0.162	0		
Position	-999979	-1280569		
(Fast)	-999978	-1282127		
(Sigma)	9	247895		
Pos. Offset	0	0		
Pos. Offset	Compute	Compute		
Pos. Scale	1000000.000	1000000.000		
Precision	0	0		

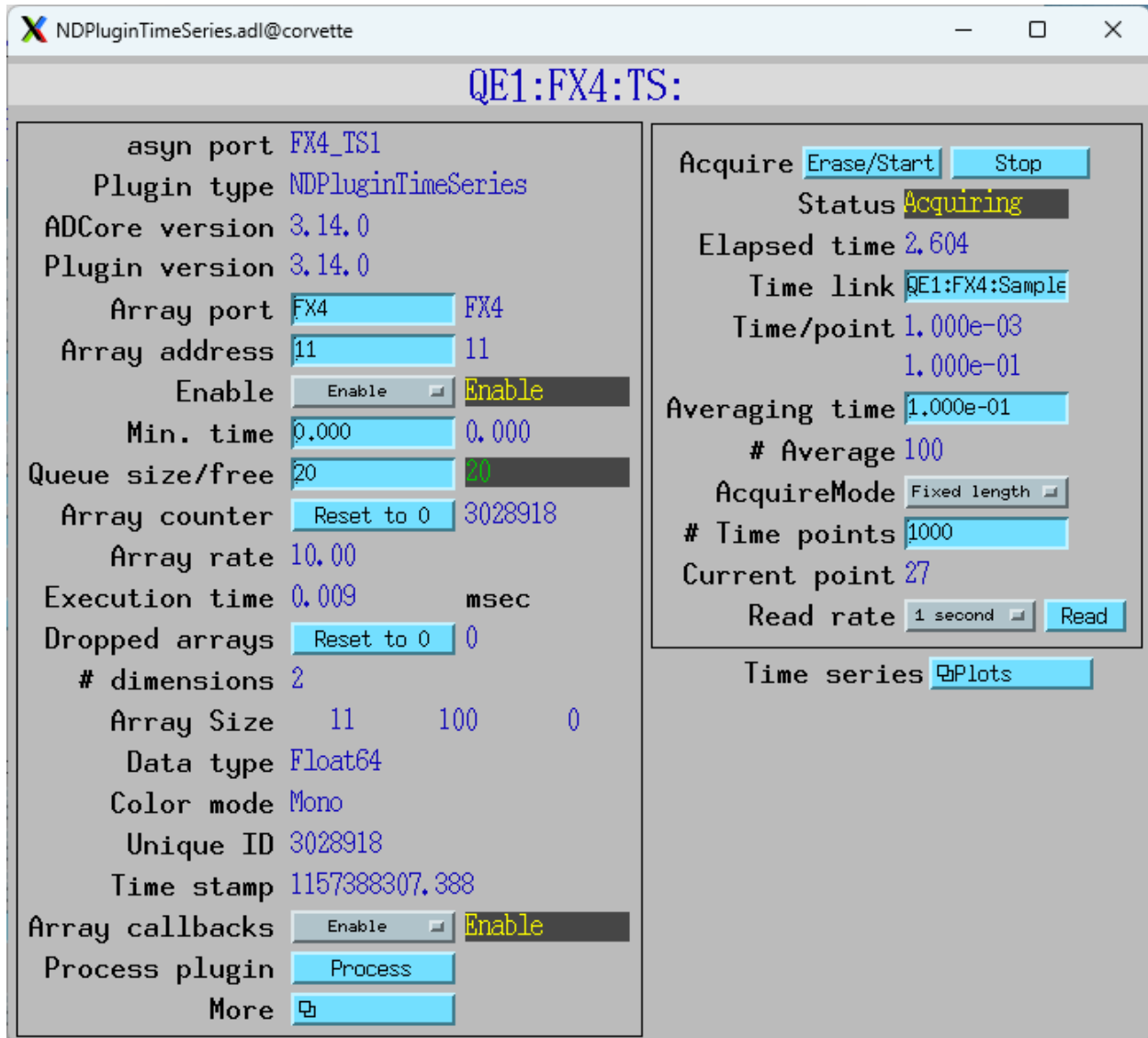
The important settings are:

- Range: 100 nA slow (1000 Hz bandpass filter)
- ValuesPerRead: 100. This means that the FX4 is doing on-board averaging for 0.001 seconds, because it digitizes at 100 kHz. This allows performing the tests in this report at up to 1000 points/s. The current report is just for 10 points/s.
- CurrentUnits: nA. This means that the values received will be in unit of nA. This is not critical, uA or pA could also be used, it just shifts the decimal point.

Note that the AveragingTime and FastAveragingTime are not important for this test. That is because we are collecting data with the TimeSeries plugin, which does its own averaging independent of AveragingTime and FastAveragingTime.

For tests on the 0-100 μ A range the only differences are Range = $\pm$ 100 uA and CurrentUnits = uA so the units are in μ A.

This is a screenshot of the TimeSeries plugin that is used to collect the data.



The important settings are:

- TSAveragingTime: 0.1. The averaging time is set 0.1 second, while the averaging time in the FX is 0.001 seconds. This result in TSNumAverage=100.
- TSAcquireMode: FixedLength. The plugin will collect the desired number of points and stop, rather than the continuous Circular Buffer mode.
- TSNumPoints: 1000 The plugin will collect 1000 time points at 0.1 second each, so 100 seconds total acquisition time.
- TSAcquire: This is set to 1 with the EraseStart button to begin data acquisition.

TetrAMM configuration

The following screenshot shows the configuration of the TetrAMM for the tests on the 0-100 nA current range

quadEM.adl@corvette

Quad Electrometer (QE1:TetrAMM:)

Model **TetrAMM**

Firmware **TETRAMM:3.3.09:IV4 120UA 120NA:HV 2000V POS**

Sample time **1.0000e-03**

Acquire **Done** **Acquire**

Acquire mode **Continuous** **Continuous**

Range **+/- 120 nA** **+/- 120 nA**

Weights Geometry **Diamond** **Diamond**

Values per reading **100** **100**

Averaging time **0.10** **0.10**

Fast averaging scan **I/O Intr**

Fast averaging time **0.050** **0.050**

Acquisitions **201** **201**

Acquisitions done **54**

Read data **Read**

To average **100**

Averaged **100**

To average fast **50**

Channels **4** **4**

Read format **Binary** **Binary**

Trigger mode **Free run** **Free run**

Trigger polarity **Positive** **Positive**

Bias state **On** **On**

Bias voltage **-1000.00** **0.00**

Bias interlock **Off** **Off**

Range 1 **+/- 120 nA** **+/- 120 nA**

Range 2 **+/- 120 nA** **+/- 120 nA**

Range 3 **+/- 120 nA** **+/- 120 nA**

Range 4 **+/- 120 nA** **+/- 120 nA**

Interlock readback **HV Enabled**

Bias readback **On**

Voltage readback **6.90**

Current readback **1.10**

Temperature **31.00**

Buffers

Pre-allocate **100** **Allocate**

Used/Alloc/Free **71** **198** **127**

Memory max/used (MB) **0.0** **4.4**

Buffer & memory polling **1 second**

Empty free list **Empty**

Signal **1** **2** **3** **4**

Current **1.009** **0.003** **0.001** **0.001**

(Fast) **1.013** **0.002** **0.001** **0.001**

(Sigma) **0.147** **0.001** **0.001** **0.001**

Offset **0.001** **-0** **0.001** **0.000**

Compute **Compute** **Compute** **Compute**

Scale **1.000e+09** **1.000e+09** **1.000e+09** **1.000e+09**

Precision **3** **3** **3** **3**

X Y All

Sum **1.015** **0.002** **1.016**

(Fast) **1.014** **0.002** **1.015**

(Sigma) **0.147** **0.001** **0.148**

Difference **-1.009** **-0.000**

(Fast) **-1.008** **-0.000**

(Sigma) **0.147** **0.001**

Position **-6105.3** **-7232.0**

(Fast) **-6102.0** **-10447.8**

(Sigma) **16.4** **67967.8**

Pos. Offset **-3843.0** **-2508.7**

Pos. Offset **Compute** **Compute**

Pos. Scale **10000.000** **10000.000**

Precision **1** **1**

Time series

Acquire **Erase/Start** **Stop** **Acquiring** **Plugin control**

Time domain plots **Frequency domain plots**

X combined **X combined**

Y combined **Y combined**

Individual **Individual**

Status

Reset **Reset**

Ring overflows **0**

Status rate **1 second**

Asyn record **Plugins**

Attributes

File

Macros

Status **File not found**

The important settings are identical to the FX4, except that Range = +/- 120 nA rather than +/- 100 nA. The settings on the TimeSeries plugin are identical to the FX4.

For tests on the 0-100 μ A range the only differences are Range = +/- 120 μ A and Current1Scale = 1e6 rather than 1e9 so the units are in μ A.

SR570, V2F100, and USB-CTR08 configuration

The following settings were used for the SR570 for the 0-100 nA tests.

- Sensitivity: 20 nA/V
- Input offset: -2 nA, set to get V/F output counts close to zero with zero current.
- Input filter: 12 dB lowpass, 1 kHz frequency

The following settings were used for the SR570 for the 0-100 μ A tests.

- Sensitivity: 20 μ A/V
- Input offset: -100 nA, set to get V/F output counts close to zero with zero current.
- Input filter: 12 dB lowpass, 1 kHz frequency
- The gain of the SR570 was as expected, a nominal 100 μ A produced 5.06 volt output.

The VSF100 was configured as follows:

- Vin range: 10 V
- Vin polarity: Positive
- Fout range: 10 MHz
- Foffset: No

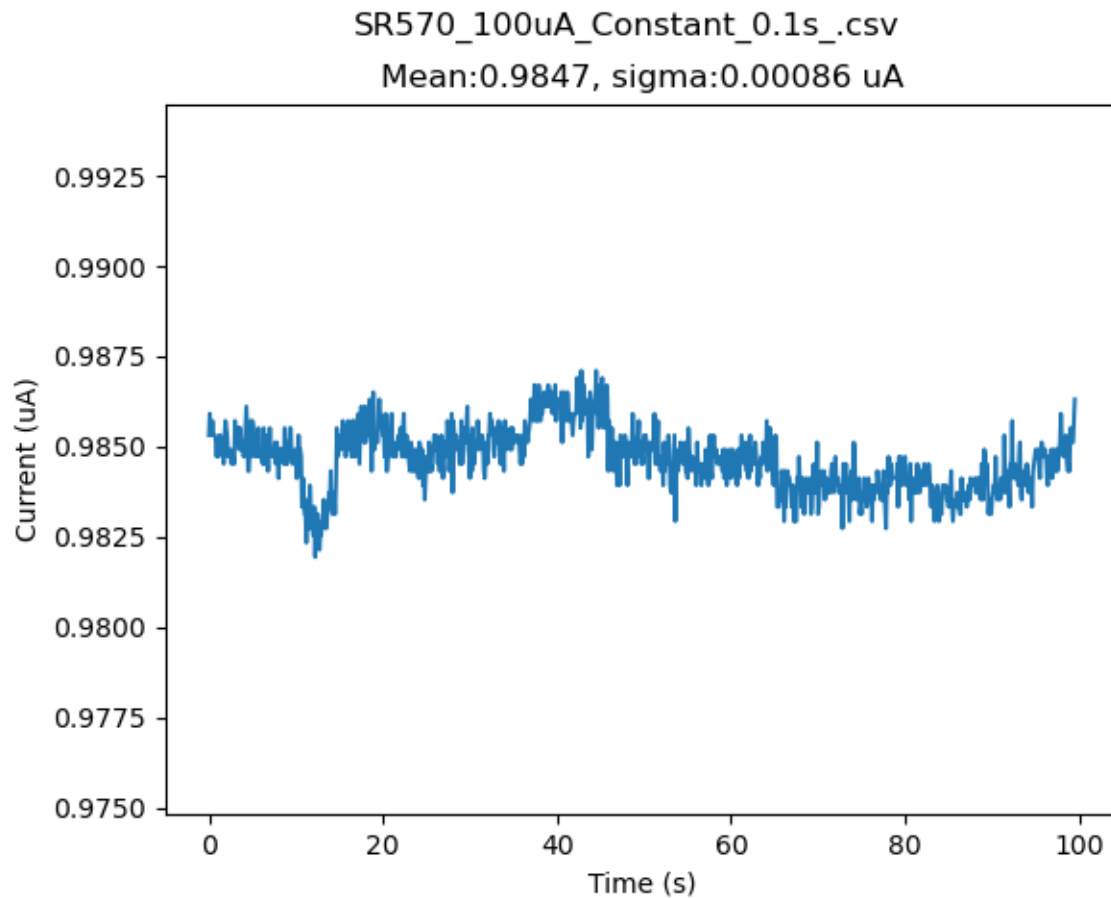
I tested 2 different V2F100 units. On the first one (serial #335) the output was not what I expected. When configured for 10V input = 10 MHz output then for a 5 V input I would have expected 5 MHz output. I actually observed 7.5 MHz output for 5V input. I then tested a second unit (serial #445) and it had the expected gain of 1 MHz/V. I used serial #012 for the measurements in this report. This is the unit normally used in the 13-ID-E station.

On the test with the constant voltage serial #335 had a standard deviation of 0.776%, while serial #445 had a standard deviation of 0.052%, 15 times lower.

Results for constant 1 μA current and 100 μA full scale

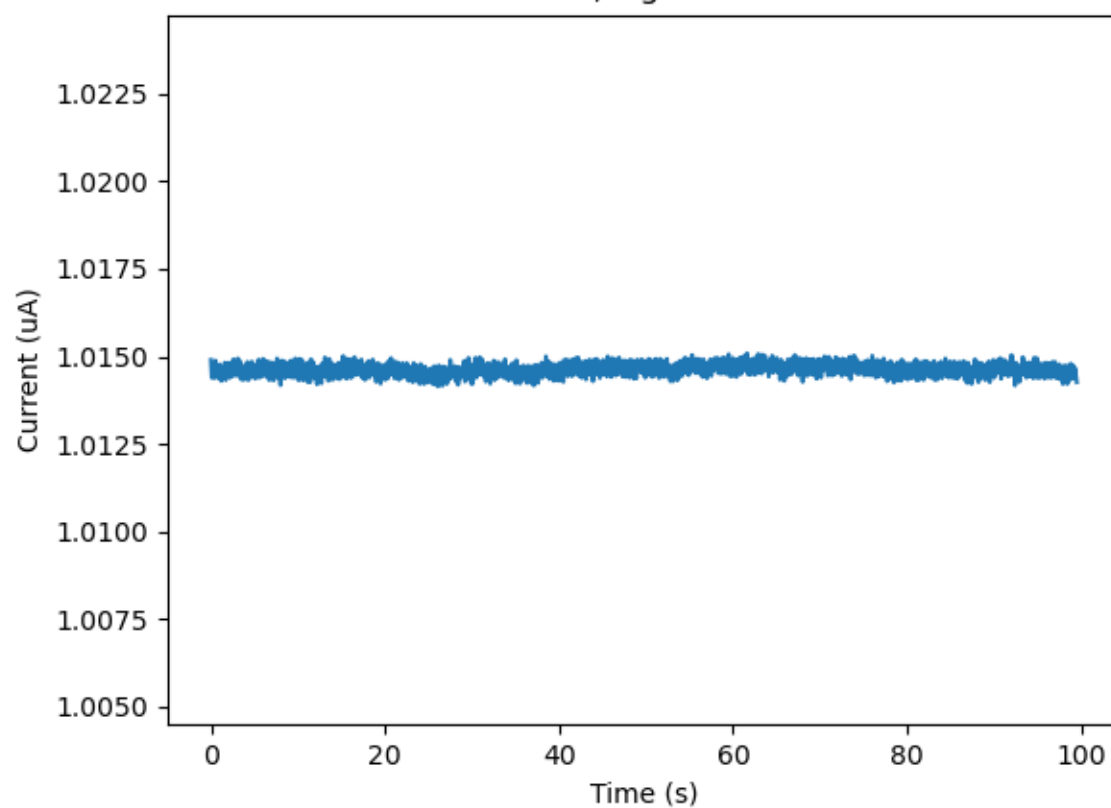
The following plots show the measured current with the vertical range being $\pm 1\%$ of the mean.

The noise on the TetrAMM is about 4 times less than the SR570. The noise on the FX4 is 26 times less than the noise on the SR570 and 6 times less than the TetrAMM.



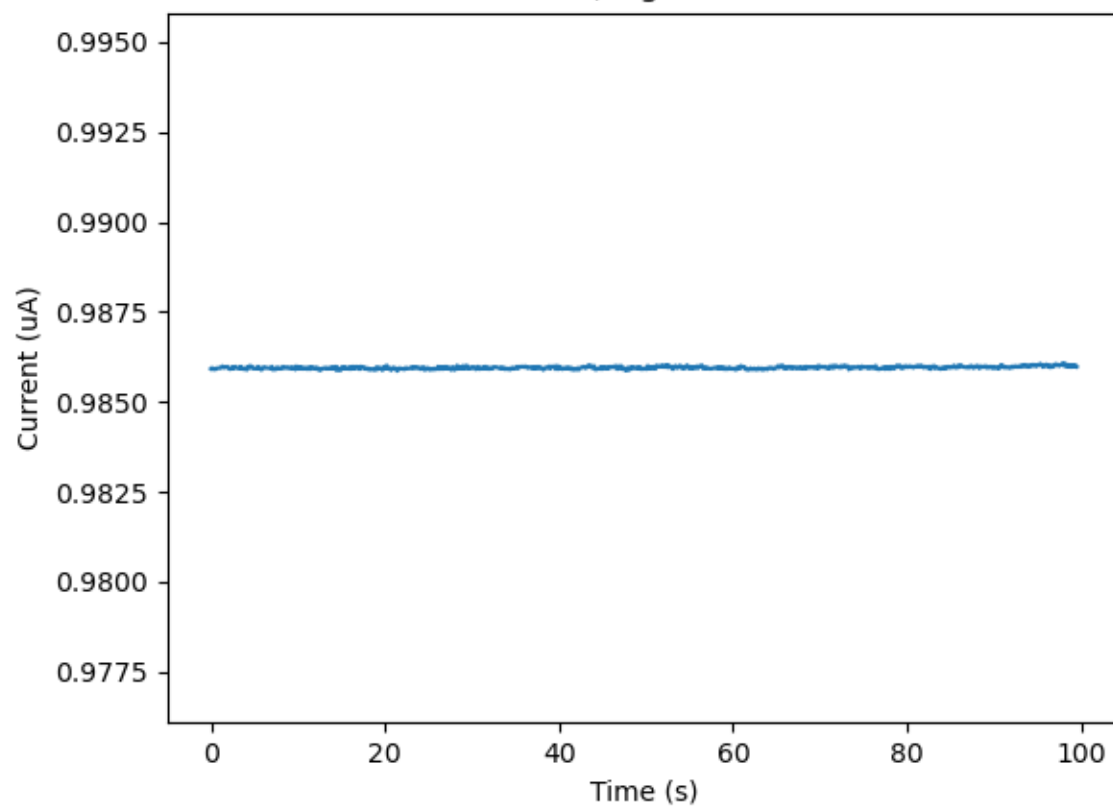
TetrAMM_100uA_Constant_0.1s_.csv

Mean:1.0146, sigma:0.00020 uA



FX4_100uA_Constant_0.1s_.csv

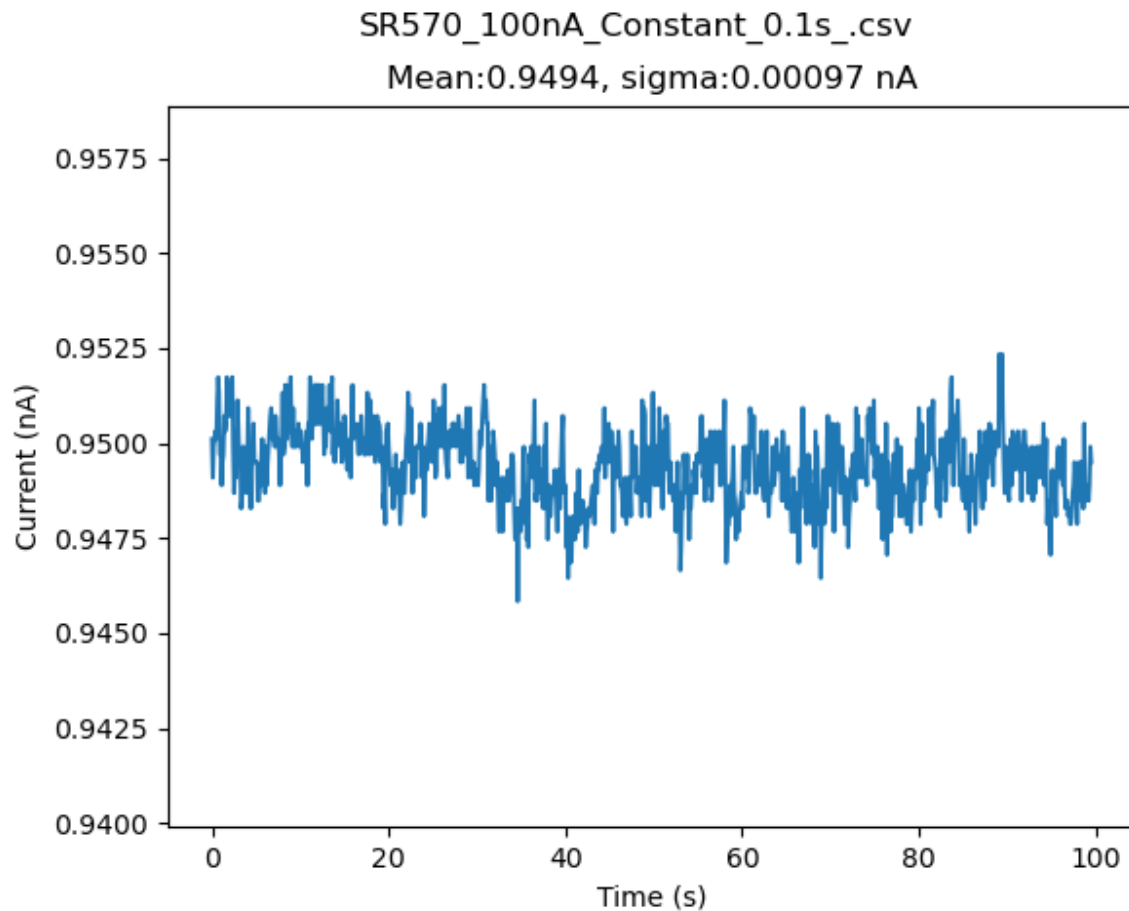
Mean:0.9860, sigma:0.00003 uA



Results for constant 1 nA current and 100 nA full scale

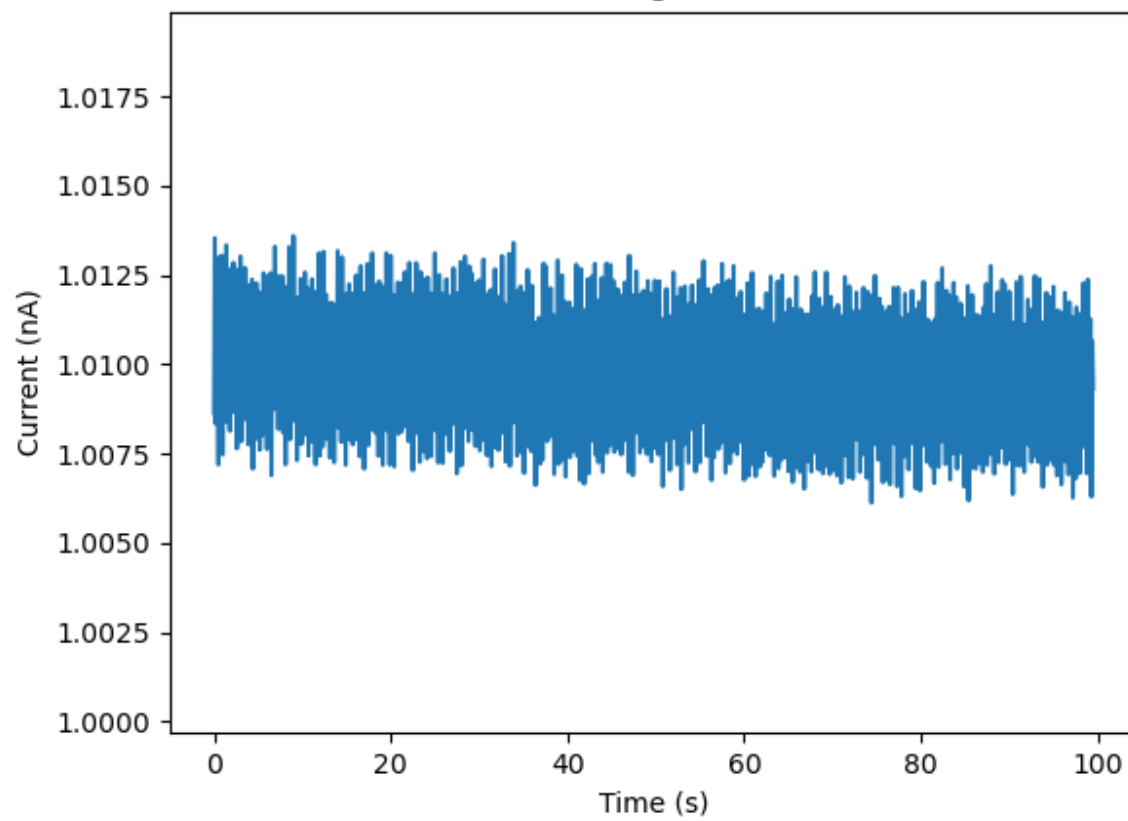
The following plots show the measured current with the vertical range being $\pm 1\%$ of the mean.

The noise on the TetrAMM is about 1.9 times greater than the SR570, and has a high-frequency component. The noise on the FX4 is 1.8 times the noise on the SR570 and has a long term drift.

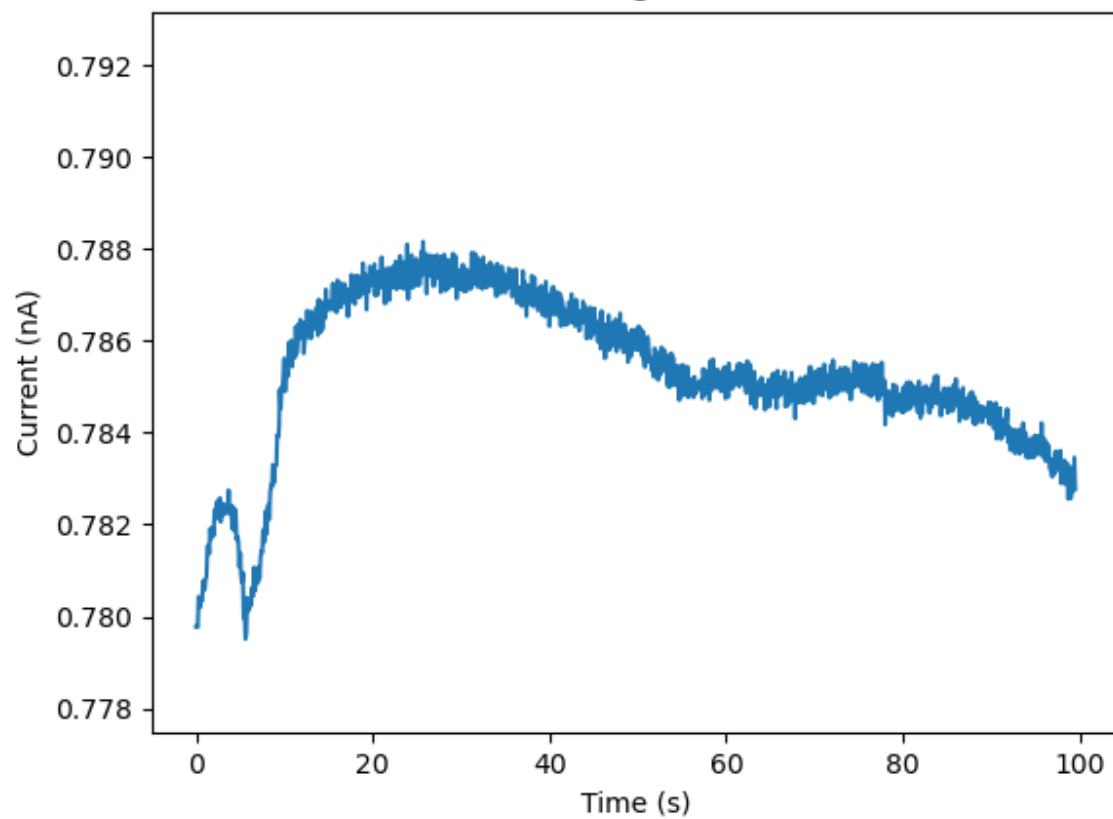


TetrAMM_100nA_Constant_0.1s_.csv

Mean:1.0098, sigma:0.00186 nA

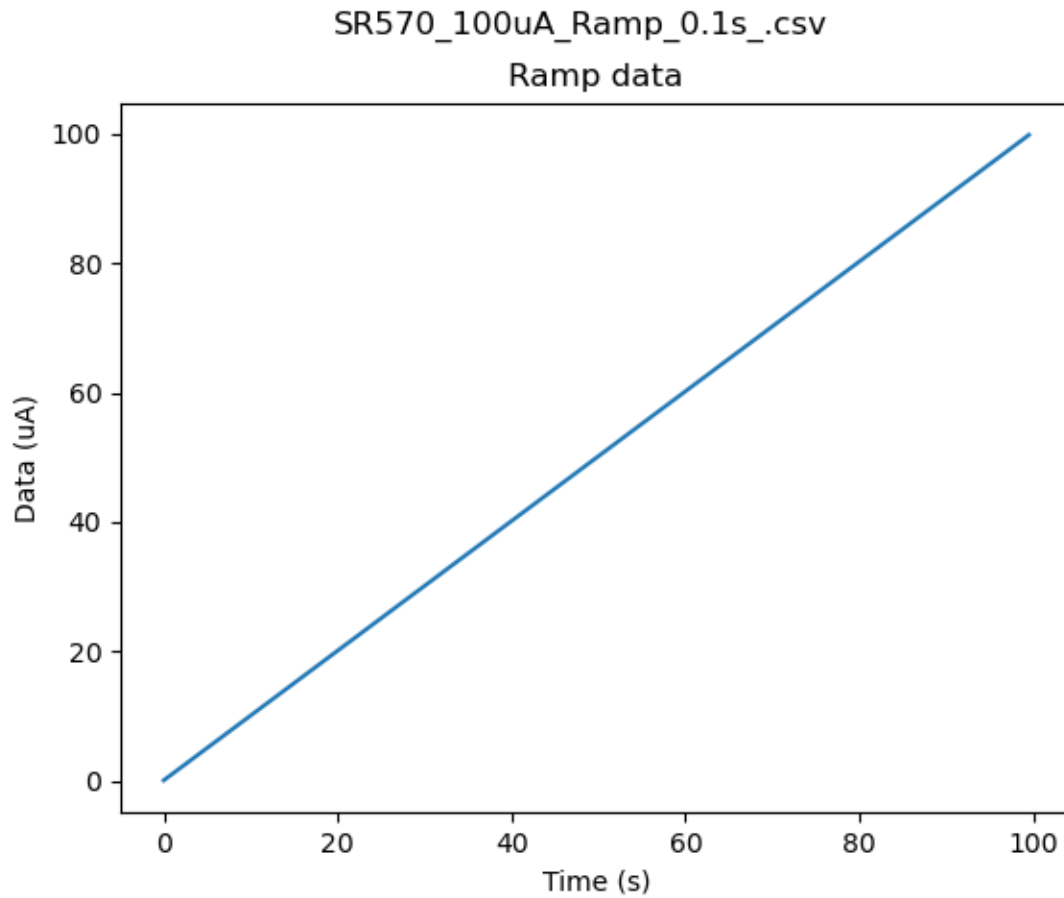


FX4_100nA_Constant_0.1s_.csv
Mean:0.7853, sigma:0.00173 nA



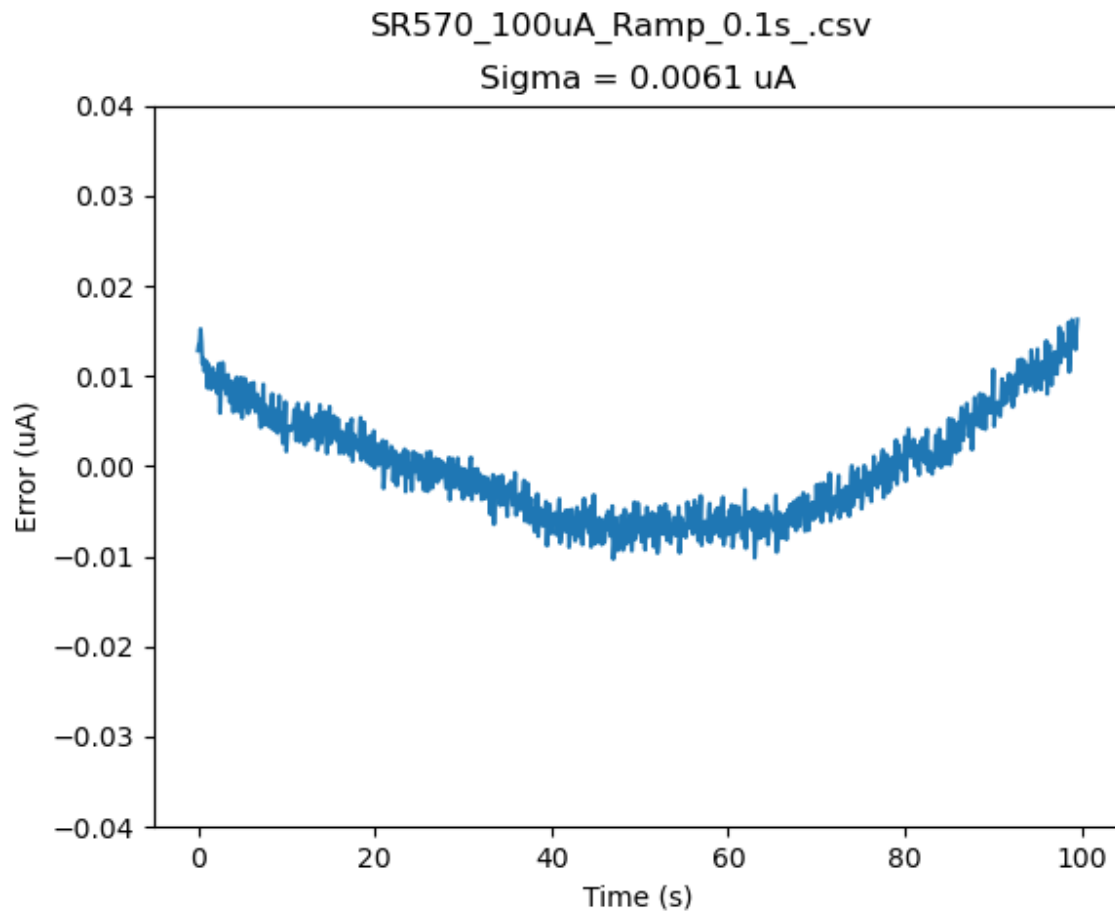
Results for linear ramp from 0 to 100 μA

The following plot shows the measured current for the linear ramp with the SR570. The plots for the TetrAMM and FX4 look identical, since the only differences are small deviations from the straight line.



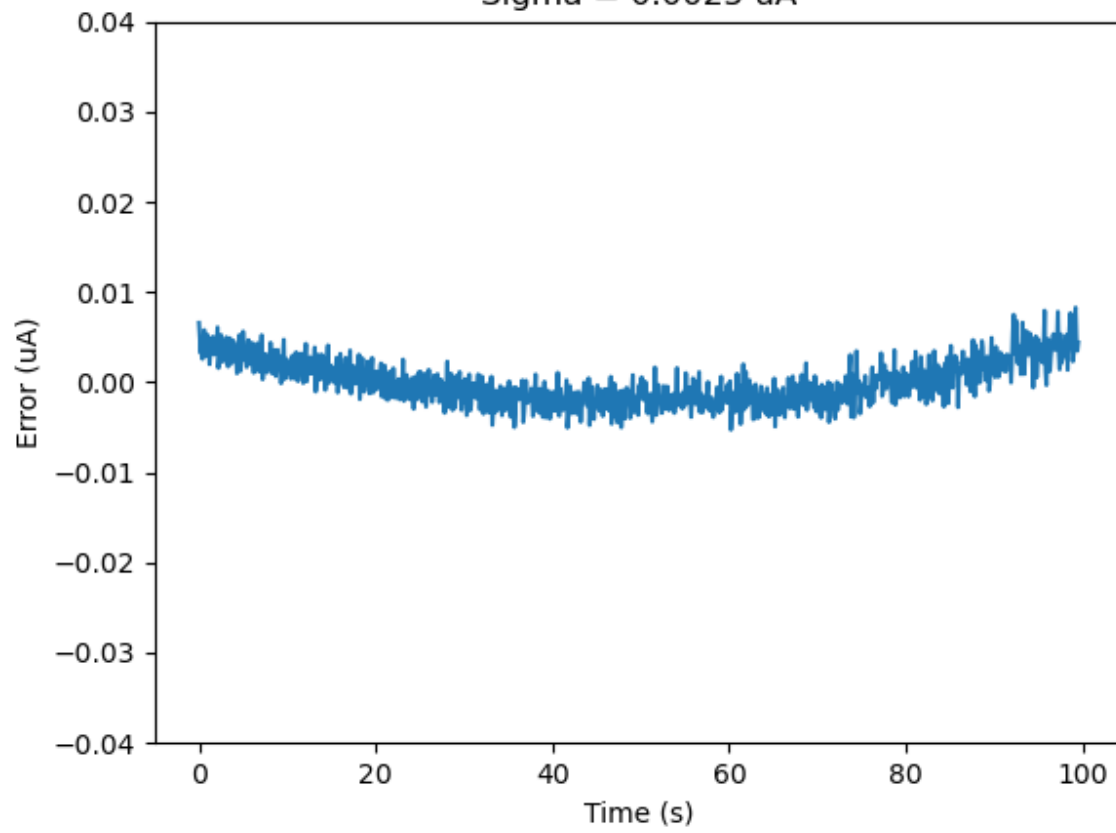
The following plots show the deviations from the best fit straight line for each device.

The error for the SR570 is 2.4 times larger than for the TetrAMM and 1.4 times larger than for the FX4.



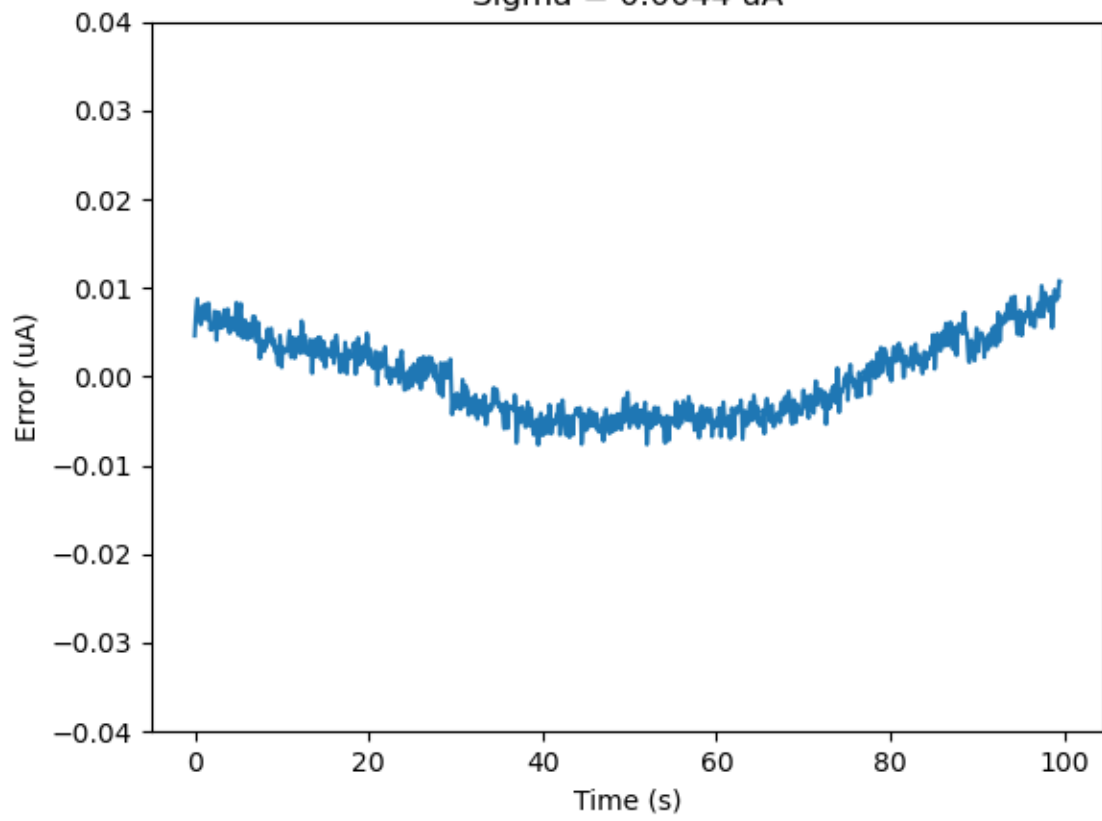
TetrAMM_100uA_Ramp_0.1s_.csv

Sigma = 0.0025 uA



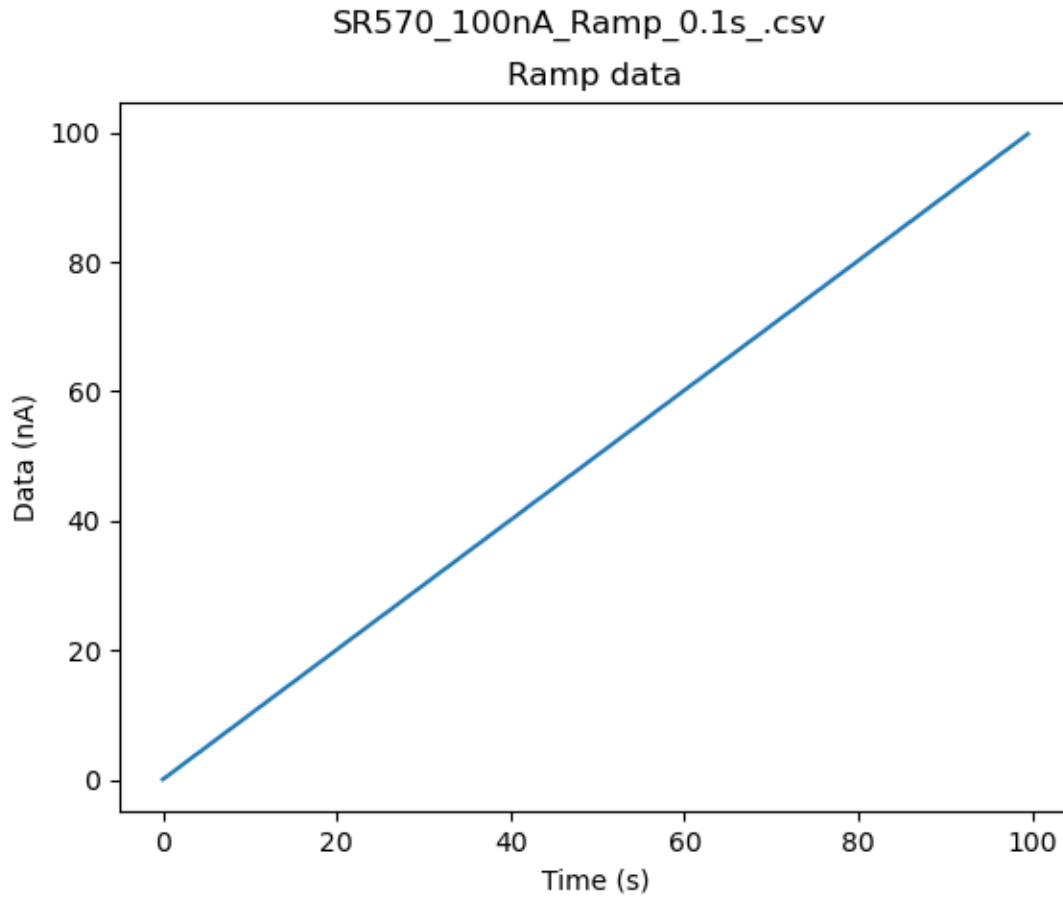
FX4_100uA_Ramp_0.1s_.csv

Sigma = 0.0044 uA



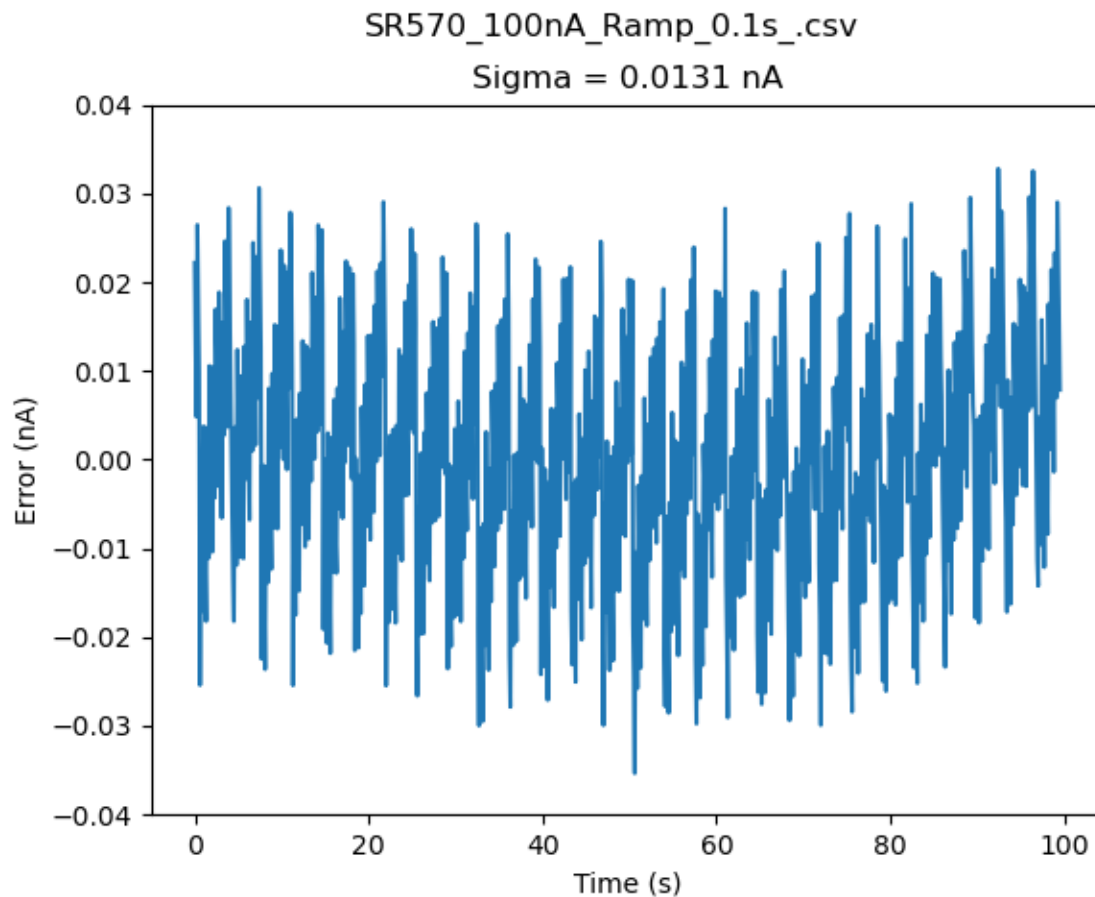
Results for linear ramp from 0 to 100 nA

The following plot shows the measured current for the linear ramp with the SR570. The plots for the TetrAMM and FX4 look identical, since the only differences are small deviations from the straight line.



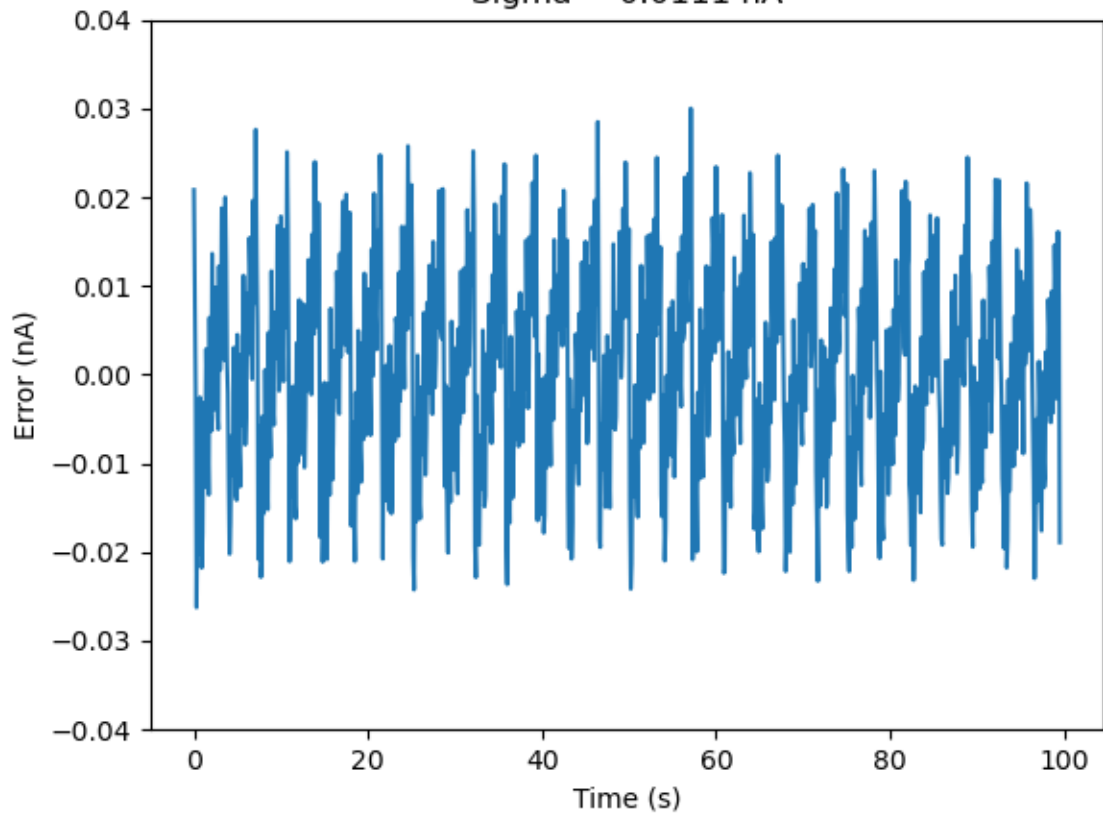
The following plots show the deviations from the best fit straight line for each device.

The errors are all very similar with sigma of ~ 0.01 nA and a distinct periodicity. I now realize the cause for this. Ideally the 100 nA signal should be generated using a 100 Mohm resistor and a 10V DAC output. However, the largest resistor we had was only 10 M ohm. This means that 100 nA requires 1 V from the DAC. The DAC on the USB-1808X is 16-bits and has a fixed range of ± 10 V. Each output bit is thus $20000 \text{ mV} / 65536 = 0.3 \text{ mV}$. The ramp is 1 V in 1000 steps, or 1 mV per step. Since the bit resolution is 0.3 mV there is up to 30% error from the desired voltage and the actual voltage. We need to fix this by either obtaining a 100 Mohm resistor, or by building a voltage divider to use with the 10 Mohm resistor.



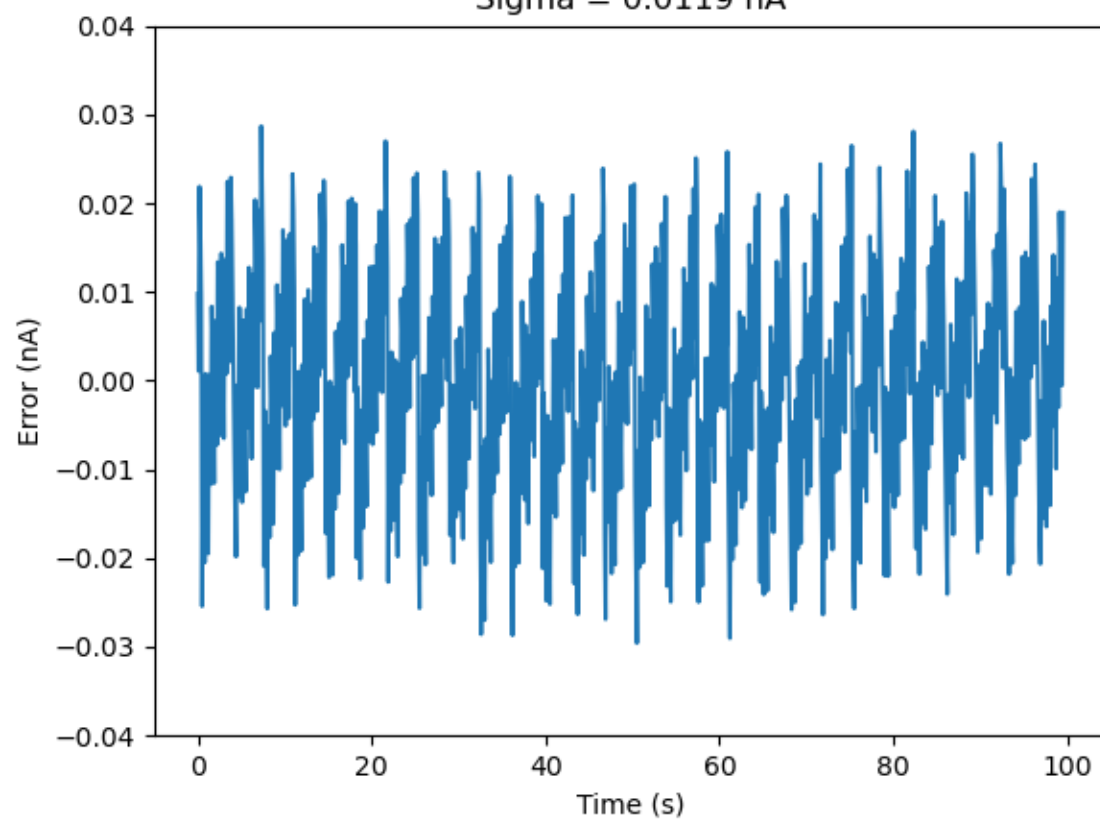
TetrAMM_100nA_Ramp_0.1s_.csv

Sigma = 0.0111 nA



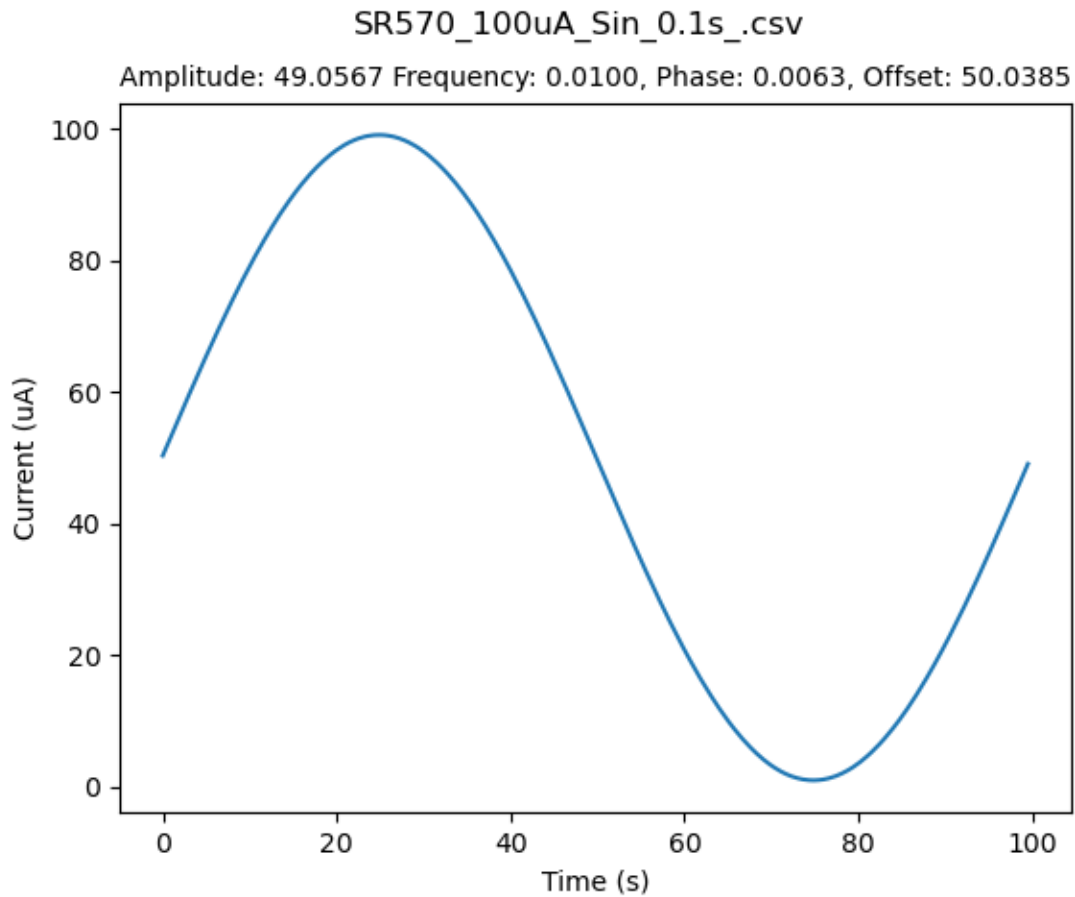
FX4_100nA_Ramp_0.1s_.csv

Sigma = 0.0119 nA



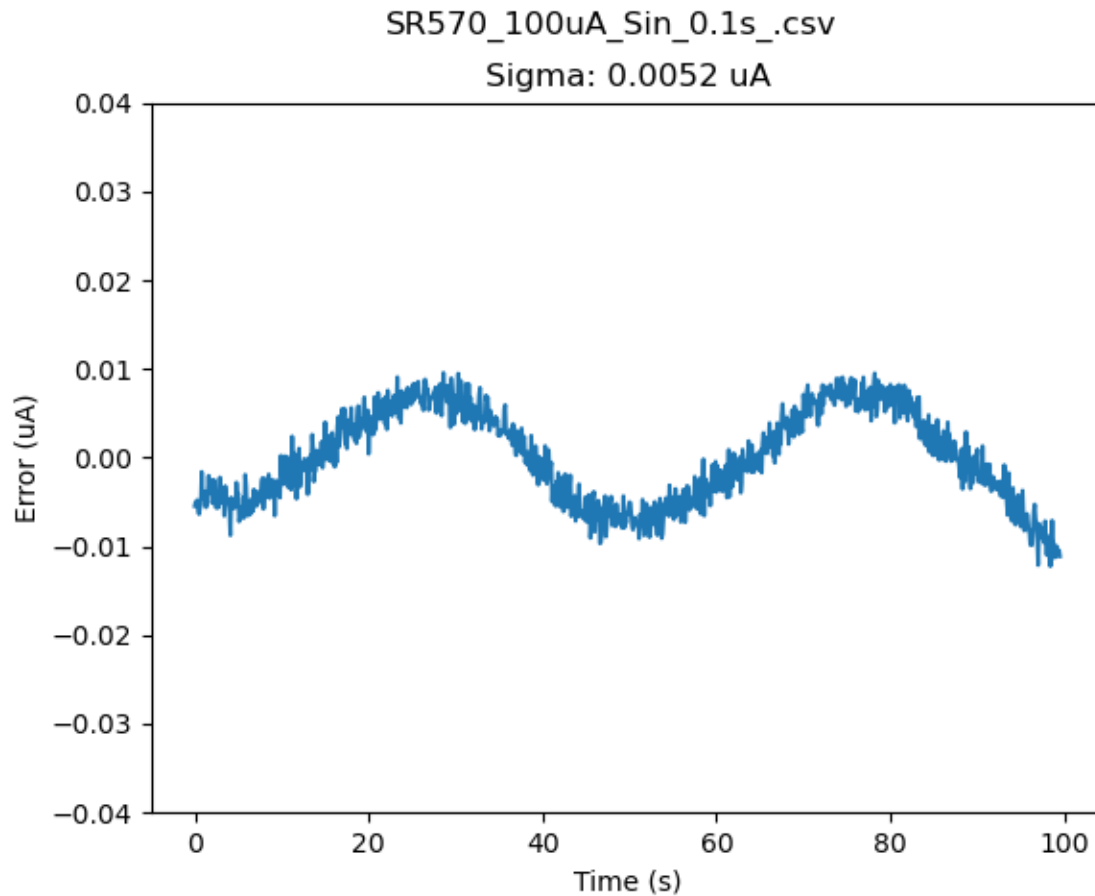
Results for sine wave from 1 to 99 μA

The following plot shows the measured current for the sine wave input with the SR570. The plots for the TetrAMM and FX4 look identical, since the only differences are small deviations from the ideal sine wave. The analysis code does a best fit to the sine wave, which when plotted on this scale is also indistinguishable from the measured data. The fit coefficients are shown in the plot title.



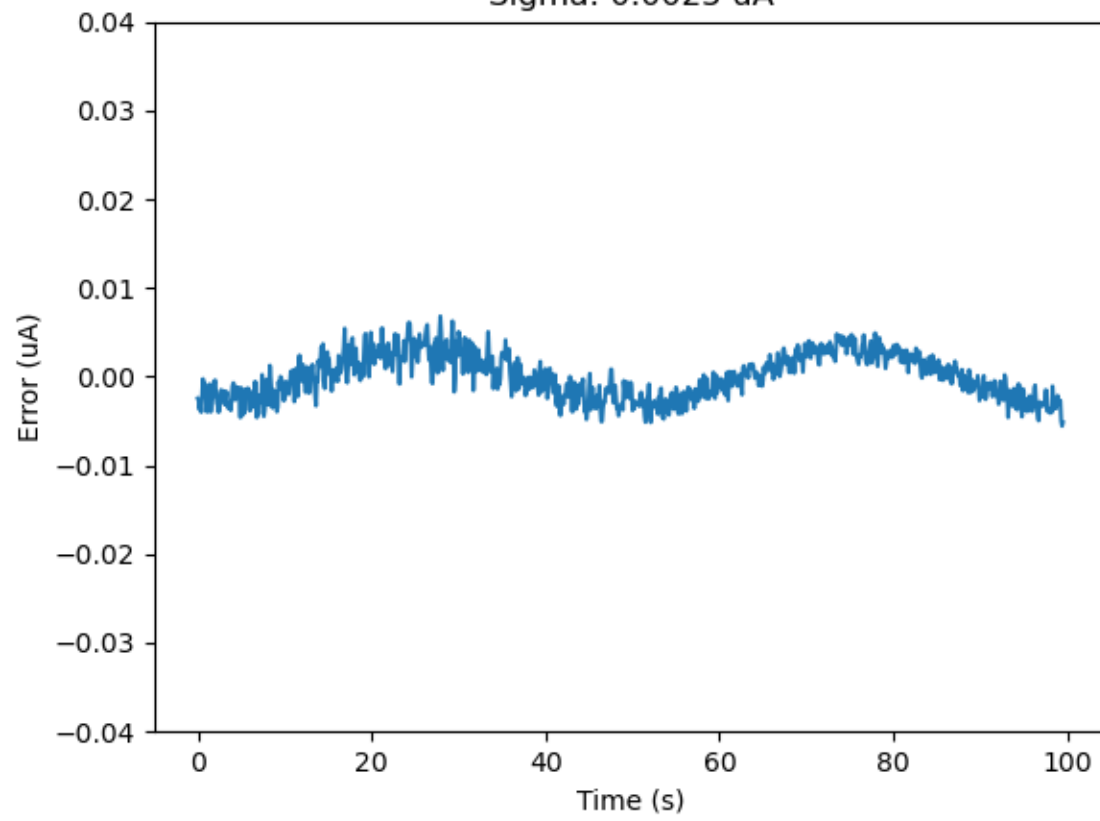
The following plots show the deviations from the best sine wave for each device.

The error for the SR570 is 2.1 times larger than for the TetrAMM and 1.2 times larger than for the FX4. The sigma values for this test are nearly identical to those for the 100 μA linear ramp test discussed above.



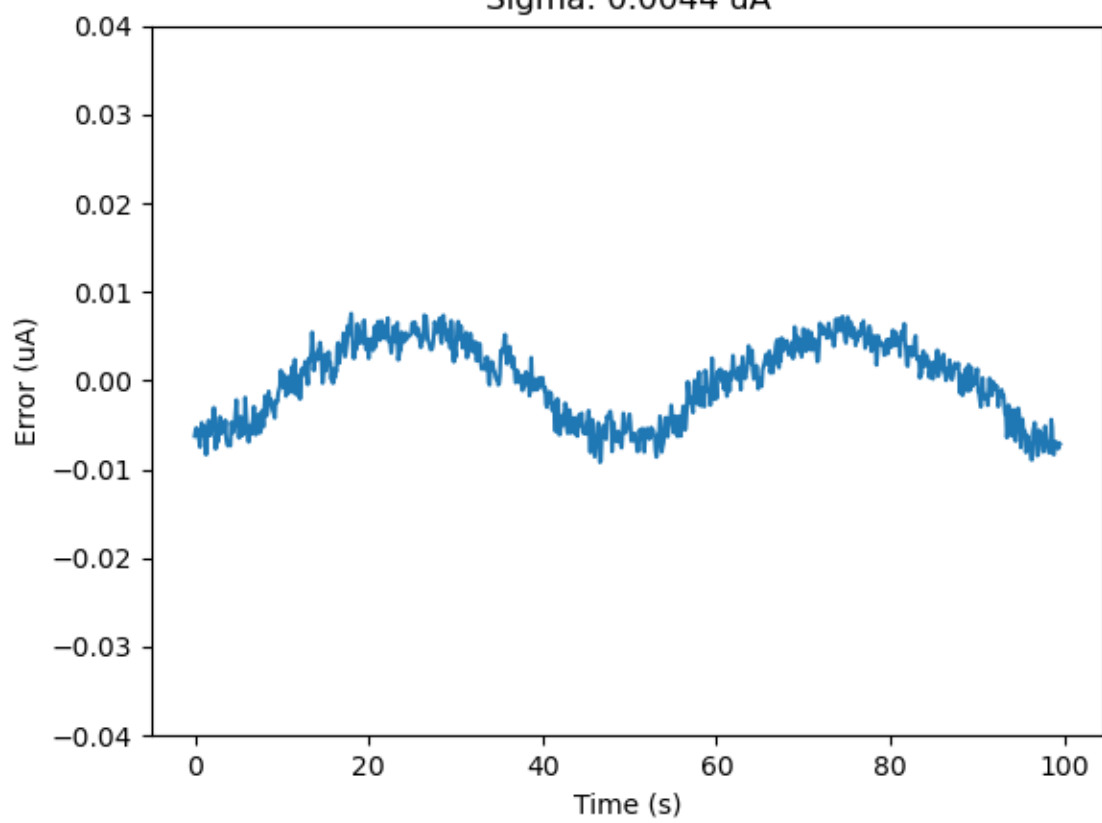
TetrAMM_100uA_Sin_0.1s_.csv

Sigma: 0.0025 uA



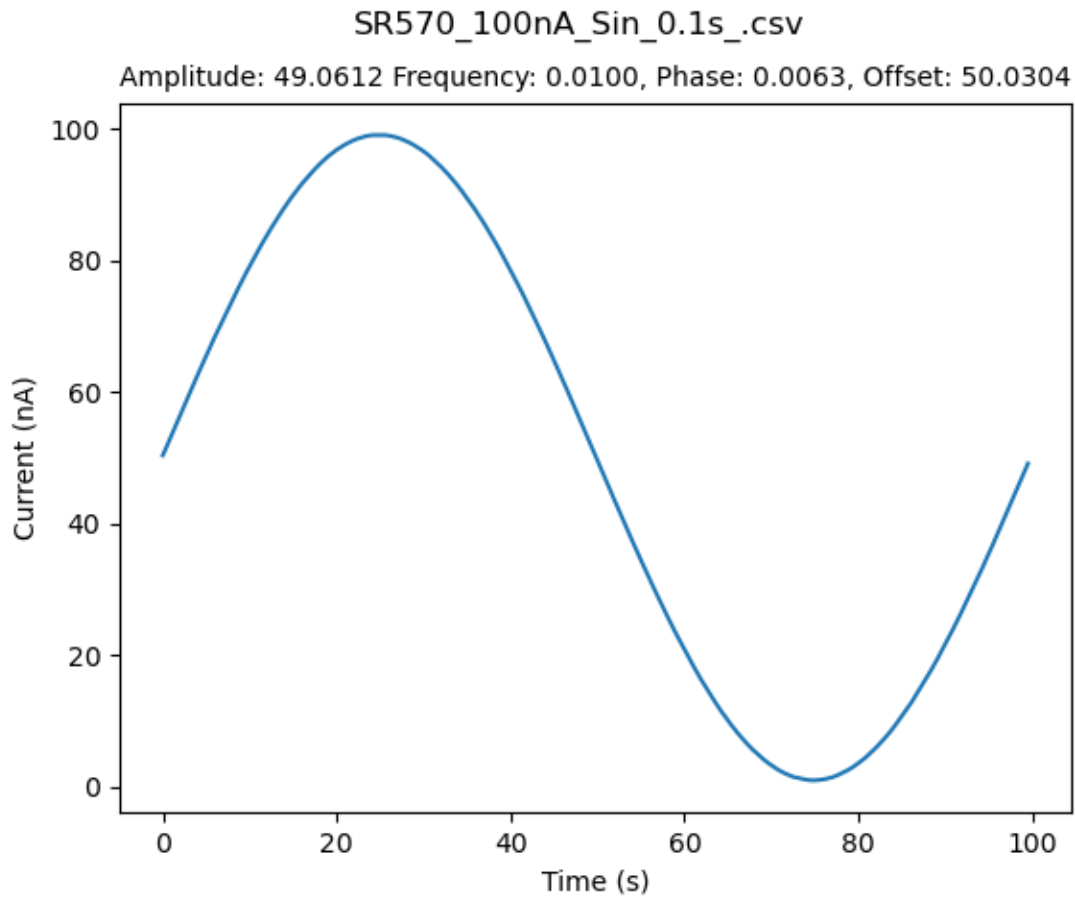
FX4_100uA_Sin_0.1s_.csv

Sigma: 0.0044 μA



Results for sine wave from 1 to 99 nA

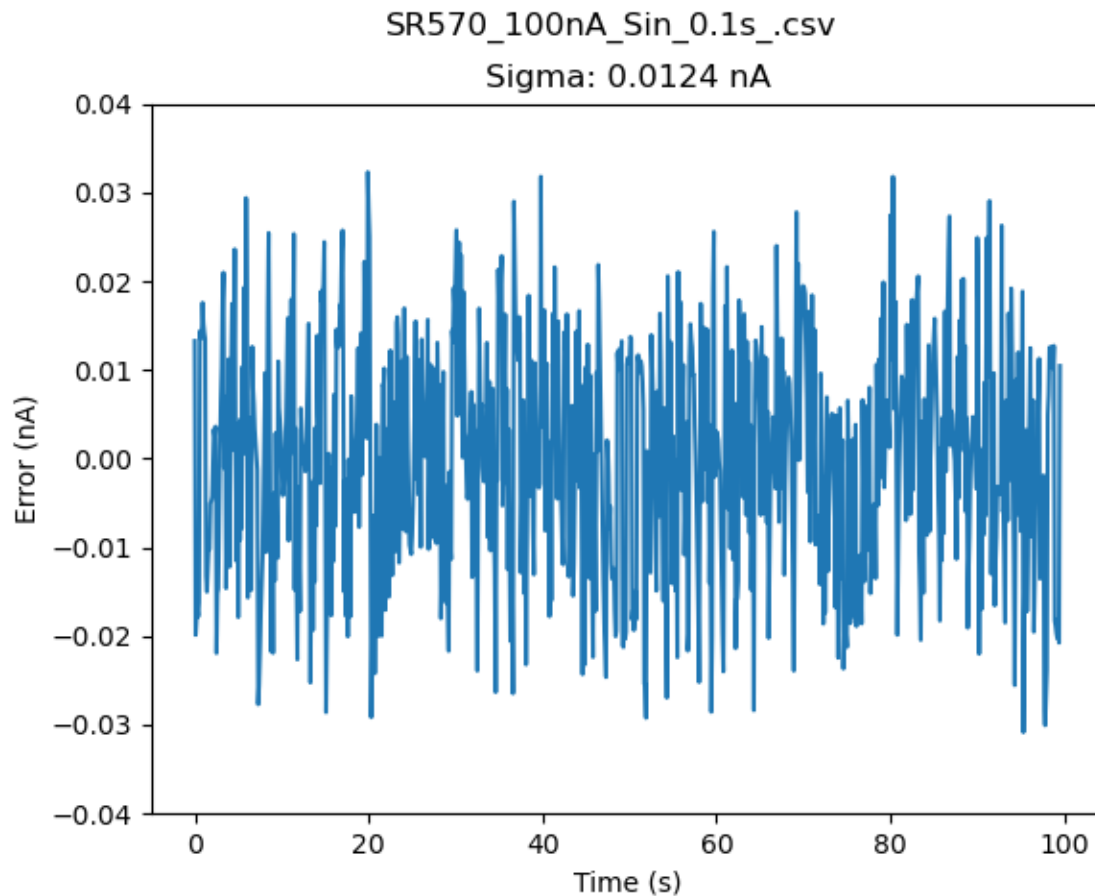
The following plot shows the measured current for the sine wave input with the SR570. The plots for the TetrAMM and FX4 look identical, since the only differences are small deviations from the ideal sine wave. The analysis code does a best fit to the sine wave, which when plotted on this scale is also indistinguishable from the measured data. The fit coefficients are shown in the plot title.



The following plots show the deviations from the best fit straight line for each device.

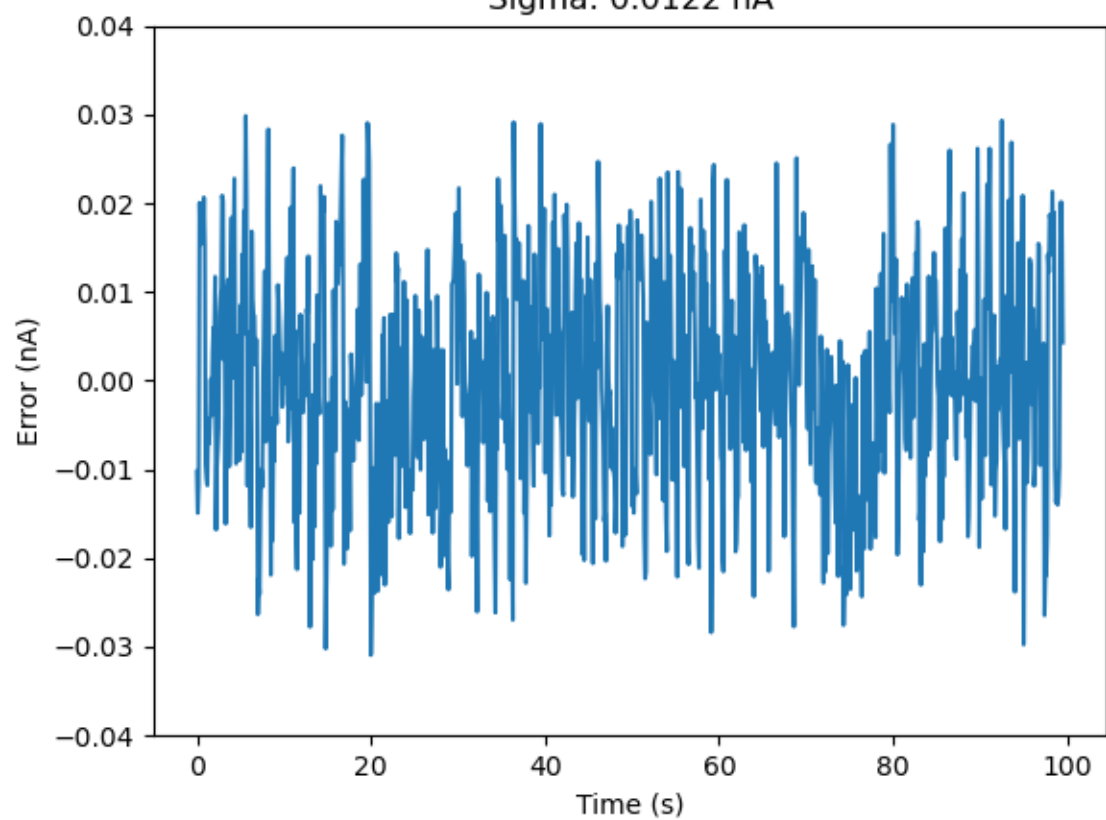
The sigma values for this test are nearly identical to those for the 100 nA linear ramp test discussed above.

The errors are all very similar with sigma of ~ 0.01 nA and a distinct periodicity. The reason for this is probably the same as for the 100 nA linear ramp test discussed above, i.e. that the precision of the DAC current output is too small because only a 0-1V output is being used out of a total range of 20 V with 16-bit precision.



TetrAMM_100nA_Sin_0.1s_.csv

Sigma: 0.0122 nA



FX4_100nA_Sin_0.1s_.csv

Sigma: 0.0107 nA

